IN A DOMAIN TWO ELEMENTS ARE ASSOCIATES EXACTLY WHEN THEY GENERATE THE SAME PRINCIPAL IDEAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Generation of a principal ideal by a multiple. Used in both directions of the counted proof (lines 29592-29594). Lets the inclusion of principal ideals be translated into a divisibility relation between the generators.
- Cancellation of a nonzero factor in a domain. Applied at proof line 29595. Turns the identity x = fg x into fg = 1, so that the two multipliers are units.

Lemma 0.1. *Let R be a domain. Let $x, y \in R$. Then x, y are associates if and only if $(x) = (y)$.*

Proof. If $x = uy$ for some unit $u \in R$, then $(x) \subset (y)$ and $y = u^{-1}x$ so also $(y) \subset (x)$. Conversely, suppose that $(x) = (y)$. Then $x = fy$ and $y = gx$ for some $f, g \in A$. Then $x = fgx$ and since R is a domain $fg = 1$. Thus x and y are associates. □

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